Accelerometer Sensor Network For Reliable Pothole Detection

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***Abstract*— This paper represents a practical fully automated solution for evaluating the road surface in a desired section. The suggested system for pothole detection uses detection node with two accelerometers. Its main goal is to make analysis of the potholes on the road surface and to visualize on the map the bigger ones, thereby increasing the safety and comfort of all passengers. The system exploits two detection methods, based on z-axis algorithms, to determine the number and location of potholes using a mobile phone and an external controller with a GY-521 sensor. When both devices detect a pothole, the sensor node sends an alert with GPS coordinates to a central node, which maintains the database. In such manner, the system response is delayed inversely with the route load, but the reliability increases significantly. The sensor network backbone takes advantage of the existing mobile network technologies. The proposed system is fully compatible with the concepts for smart cities and easy can become part of their environment.**

Keywords*— *pothole detection, sensor network, accelerometer, z-axis algorithms, threshold value, GPS position, map visualization

I. INTRODUCTION

The development of the electronic components allowed the integration of thermometers, multi-axis gyroscopes and accelerometers into miniature chip housings, with a wide range of applications in industry. Most smartphones, some vehicles, some security devices, drones, toys and game controllers use such a chip. It recognizes the position of the device in space and allows it to react adequately to changes in this position.

Due to its capabilities, one of its first applications was for early warning of elderly people falling to the ground. It implements a plan to reduce injuries related to falls and prevent their consequences, thus reducing the risk of prolonged periods of lying down. Typical fall detection systems determine whether there has been a fall by analyzing patterns of human movement, such as changes in speed or changes in body posture. Many authors [1,2] describe a drop detection algorithm based on predefined threshold values from the sensors. The total acceleration vector A_{cc}, containing both static and dynamic components of the acceleration, is calculated from sample data using the following formula:

$$A_{cc} = \sqrt{(A_x)^2 + (A_y)^2 + (A_z)^2}, \quad (1)$$

where A_x, A_y and A_z represent the accelerations (g) on the axis X, Y and Z of the accelerometer.

Methods for identifying potholes are basically visual or manual. Manual identification is usually done by RIA (Road Infrastructure Agency)inspectors and in most cases is time consuming, expensive and depends on the human factor. To effectively deal with these difficulties, various automated methods for detecting potholes have been developed by various transport agencies and researchers. The most common method uses image processing. Using two-dimensional or three-dimensional images, the depth of the potholes can be determined [3,4].

There are several automotive sensor systems for detecting potholes. Some of them use accelerometers from which they collect data. Such a system is described by Eriksson et al., (2008) and is called Pothole Patrol. It uses a GPS module and a smartphone accelerometer that is attached to the dashboard of the car. The study proposes a Z-peak algorithm that takes into account the peak acceleration along the z axis as a major sign of significant road surface anomalies. The xz ratio is also taken into account. Acceleration along the x-axis can help to identify road anomalies that span the width of the road and therefore affect equally on both sides of the car. It turns out that the Z-peak algorithm gives correct results only when the speed is greater than 30 km/h [5].

Another system is Nericell proposed by Mohan, Padmanabhan and Ramjee [6] uses a z-sus pothole detection algorithm for speeds less than 25km/h. Zoysa et al., (2007) offer a system called "BusNet", which monitors environmental pollution and the condition of the road surface. In this system, the smartphones are placed in public buses [7]. The systems listed above do not require reorientation of the accelerometers built into the phones with respect to the axes of the vehicle, but there are some studies with such requirements. [7,8].

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There are also systems with advanced capabilities in which a machine self-learning algorithm is used in parallel with the pothole detection methods.

Another article that formulates different algorithms for real-time pothole detection using Android smartphones has been proposed by Artis Mednis al.[9] It describes that the results are influenced by various factors, such as vehicle speed, pothole area and some others.

There are also some other suggestions for pothole detections, but for various reasons they have failed to establish themselves in practice. Some reports describe devices that use an ultrasound sensors positioned under the vehicle. These sensors monitor the distance between a vehicle body and the road surface. Unfortunately, this type of sensor can be easily affected by dust, mud and in many cases can be destroyed mechanically.

Some other authors suggest the use of cameras. In this case, the problems can be just the same like with ultrasound sensors. In addition, image and video processing requires much more hardware computing resources.

There are several main disadvantages of the systems listed above:

- The alert threshold sensitivity of the systems depend on the weight, the suspension and shock absorbers of the vehicle;
- In some cases systems can be affected by passengers movements in the vehicle;
- Different positions of the sensors on the car body can provide a different output results;
- Strong side winds, as well as sudden starts and stops of the vehicle can cause false pothole alerts;
- Most of the suggested solutions are isolated within a single vehicle as an input source.

All listed above, show that new approaches need to be found to increase the accuracy and reliability of detection.

II. Basic Pothole Detection Methods

One of the simplest detection algorithms is Z-Thresh[12] and it is similar to the Z-Peak algorithm used in Pothole Patrol. Both of them operate with simple threshold-based heuristic. Typically, Z-Thresh is used to detect real-time events using limited hardware and software resources. In this method, it is assumed that the initial value of the accelerometer position along the Z-axis is known. The algorithm identifies changes in the amplitude of the Z-axis acceleration that exceed a predefined upper threshold value. This threshold also determines the type of roughness - if the pothole is small, the sensor will generate a value less than the threshold and no reporting event will be generated. Figure 1 shows the operation of the Z-Thresh algorithm, which measures values and generates an event when the set threshold level is exceeded.

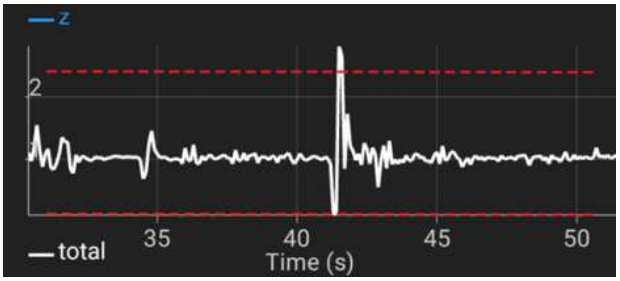


Fig.1 Z-Thresh detection event

The second most used algorithm for pothole detection is the advanced Z-Diff algorithm. It makes two consecutive measurements of the vertical acceleration along the Z axis, within a certain time slot, and analyzes the difference between their values. If this difference exceeds a predefined threshold value, a reporting event is generated. Thus, the changes in the vertical acceleration are analyzed, and not its instantaneous values, which can be easily influenced by external factors. Figure 2 shows the occurrence of a reporting event from the Z-Diff algorithm.

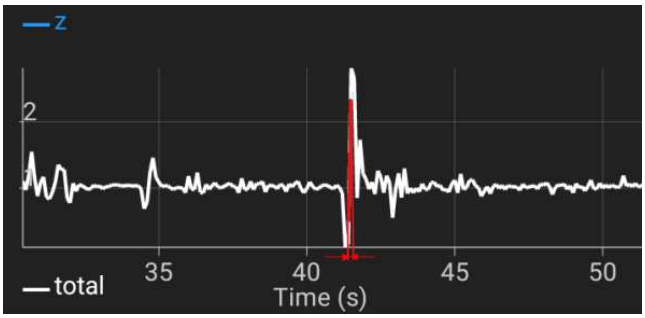


Fig.2 The Z-Diff Algorithm

In a number of studies, the external triaxial accelerometer is calibrated using the offset method[10]. The difference between the specified output value and the actual output value is designated as Offset. In a perfect case, the values of the accelerations of the x, y and z-axes should be 0, 0 and 9.81 m / s2 respectively. Here 9.81 m / s2 or 1g represents the acceleration by gravity. The difference between the values of the offset and the ideal acceleration is known as the offset error. The results obtained after calibration under ideal conditions must be equal to zero.

In the study of Schoeffler [11] accomplished with module GY-521, the accelerations of the values of x, y and z from the accelerometer are effectively observed. After observation the values of z, when the accelerometer is placed on a desk and does not move, an average value of 16300 is established. It is called baseline (acceleration of 1g). When the accelerometer is shaken, the values change significantly depending on the explored time slot, the applied movements and some other factors. The current difference from the baseline is determined as follows:

$$diff = |\ (16300 - |A_cZ|)\ |\ , \qquad (2)$$

where:

diff is the current difference from the baseline,
A_cZ is the current value of the sensor.

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III. Details Of The Suggested System

Pothole detection in our network realizes in sensor nodes that combine two separate devices – a wireless controller with gyroscope/acceleration sensor and a mobile phone with integrated sensor and positioning module inside. To achieve high reliability, a combination of two algorithms - Z-Thresh and Z-Diff is used. The external controller uses Z-Thresh, while the smartphone uses Z-Diff. The mobile phone serves also as a gateway to the Internet and tracker.

Double pothole detection with both devices in a single detection node is the first innovation that aims to increase the reliability of the whole system. Usage of both algorithms assures additional accuracy.

The external controller with the sensor is connected to the smartphone via wireless connection – Wi-Fi or Bluetooth. In our prototype it is Wi-Fi, but Bluetooth is also a good choice.

The architecture of the system is depicted on Figure 3:

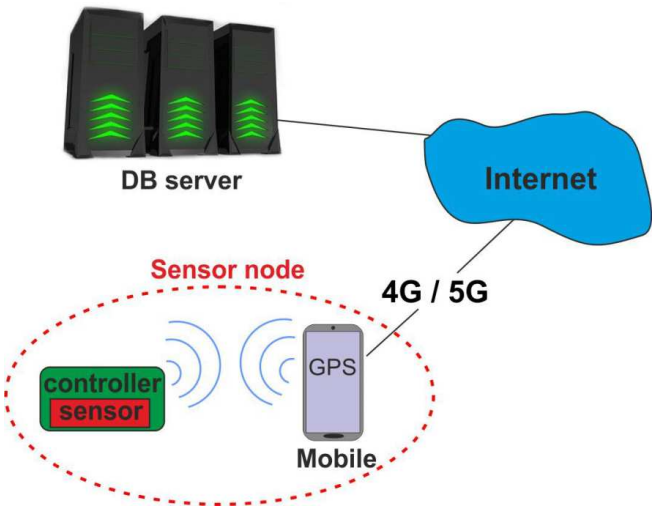


Fig.3 Pothole detection system

After receiving the result from the wireless network, the mobile phone reads its own sensor and if both results (algorithms) show that there was a pothole on the road, it sends an alert to the database server via its mobile network Internet connection. The alert contains also the GPS – coordinates that are read from the positioning module of the phone.

The basic criteria by which the parts for the prototype are selected include:

- accuracy;
- accessibility;
- possibility to work in different conditions;
- popularity in such developments.

To detect potholes through the sensors GY-521, the sensor in smartphone Samsung Galaxy J2 Pro or BMA2x2 in Nokia 5.3, z-axis acceleration analysis algorithms are used. One of the most important parts are the sensors for the external wireless controller. The conventional shock sensors are not applicable in this case because they cannot discriminate the shocks direction and will create a lot fake alerts. GY-521 is a good choice. It fully covers all requirements, listed above. It is small, cheap, easy to use and accurate. These makes it very popular for pothole detection.

NodeMCU (ESP8266) is selected wireless controller. It uses a WiFi connection and has ready libraries to work with GY-521 sensor. Another possible option is ESP32. It can operate by Wi-Fi or Bluetooth (4.2 or BLE) connection. ESP32 has also more processing resources (dual core, higher frequency, more scalable ADC and others). However, in its role as a final unit in the system, the controller does not need additional resources and therefore NodeMCU fully satisfies the requirements in this case. Both sensor and wireless controller can be seen on Figure 4.



Fig.4 External wireless controller with GY-521 sensor

In our study, the box with the controller and sensor was placed on the car floor. While driving, every time the car goes through a pothole, the results are recorded. From the collected data it is established that every time there is a current difference from the baseline - 7000 or more, it goes through a pothole and the location is recorded. In other words, this is the threshold value of the baseline. The proposed system works on Android-based smartphones. An application has been developed that reads the accelerations along the z axis from the built-in accelerometer in the phone and from the accelerometer of the external sensor GY-521.

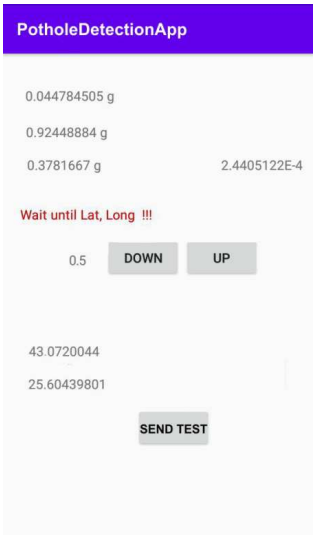


Fig.5 Interface of the developed application – the data from the smartphone can be seen on the screen

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When the set thresholds are exceeded and a pothole warnings are received from the used algorithms, the application sends GPS coordinates of the location to the Firebase Realtime Database. The detected coordinates are visualized using the Google Maps API, which provides an easy way to mark the locations of potholes on a road surface on a map and integrate it into a web page. Many markers are very suitable for showing several locations of holes in one map. Thus, the generated dynamic markers, corresponding to the number of records, indicate the locations of the holes on the Google map, using the data stored in the Firebase database.

When the database server receives an alert from a specific sensor node (vehicle), it writes the data into the database. If another alerts makes a new records for the point with the same geographical coordinates for a defined period, (depending on the traffic load of the specified road section - for example for one day), the integrated cycle algorithm marks the position on the pre-defined map. The number of matching alerts before preview on the map can be tuned as an intersection point between response time and desired degree of reliability. If a single record remains isolated and unconfirmed by others, it is deleted from the database after the specified period. This is another innovation that increases the reliability of the system.

IV. Observed Results

The proposed system for detecting potholes on the road surface has been tested in the city of Veliko Tarnovo, Bulgaria. A random route with a length of 5.00 km was chosen for the test experiment. It contained 82 potholes identified by visual inspection (images). This total number of potholes includes large potholes, small potholes, clusters of potholes, gaps and road shafts. The technical equipment used during the test experiment includes a car Seat Ibiza III, two different smartphones - Samsung Galaxy J2 Pro with 3-axis accelerometer STM LIS2HH12, Nokia 5.3 with 3-axis accelerometer Bosch BMA2x2 and a NODMCU controller with a GY-521 sensor.

In order to study the accuracy of the accelerations, the accelerometer results of each sensor unit are taken into account and analyzed. The number of holes detected by different combinations of algorithms with changing threshold values was analyzed and the combination that gives the optimal output for both accelerometers was identified. When four or more identical coordinates accumulate in the database, the hole is displayed as a marker on the Google Maps Road Map. The percentage of true and false positive results of the method, based on a smartphone and an external controller with a GY-521 sensor, is estimated using data obtained from the accelerometers of the two sensors, which are obtained synchronously.

The accelerometer results are compared with video data or images in a small number of reports, one of which is by Lekshmipathy, Velayudhan, and Mathew. The methodology used to classify elements as genuinely positive if alerts are detected within a radius of 15m from each ground truth item. Each ground truth item is classified as a real hit if, during 8 laps test drive, at least 4 alerts in different laps are detected within a radius of ≤15 m. The percentage of true and false results is obtained from the number of event locations registered repeatedly by the technical equipment in the database, but the results are also validated with images from the road section. In the performed experiments, the location of the recorded alerts, less than four times in the database corresponded to the false positive results - images from the road section in which no potholes were found.

When setting up the equipment, for using Z-Thresh algorithms, threshold values in the range between 0.1 and 1.0g were tested. For optimal Z-axis detection was chosen 0.2g, because at this value 75.60% of all detected events were defined as real hits. Thresholds between 0.1g and 1.0g were used for the Z-Diff algorithm. After further analysis, 0.2g was chosen for the optimal threshold. Using the combination of the two algorithms, 76 potholes were correctly detected, corresponding to 92% true positive results, of which 62 were detected by the Z-Thresh algorithm.

After multiple passing of the route, the proposed system displays on the road map only the large and medium-sized potholes, the location of which is registered four or more times in the database. With the applied threshold value of 0.3g for the Z-Diff algorithm used in the mobile phone together with the threshold value of 0.3g for the Z-Thresh algorithm used in the NodeMCU, 30 large and medium-sized potholes along the route were reported, 26 of which are true positive. For the Z-Thresh algorithm running on NodeMCU with the connected acceleration sensor GY-521, a threshold raw value of 7000 units was selected, which corresponds to 0.43g. Subsequent downward corrections resulted in an optimal result at a threshold of 0.3g.

The developed system is set up to mark automatically the geographical position of large and small potholes on a Google map, as shown on Figure 6.



Fig.6 Map of Veliko Tarnovo showing 26 discovered large and small potholes with the highest degree of accuracy.

When using threshold 0.8g for both algorithms, twice fewer potholes were located, taking into account only the larger 9 potholes. Figure 7 shows the 8 already discovered potholes - true positive results from multiple tests along the same route, using threshold values of 0.8g.

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Fig.7 Map of Veliko Tarnovo showing 8 large potholes found at threshold 0.8g

By combining both algorithms with the thus selected 0.3g thresholds, the system achieved an improvement in the accuracy of the results in detecting small and large holes. When setting higher thresholds, the used algorithms report fewer potholes - some of those already found. Respectively, the setting of the system with lower thresholds is not appropriate due to the registration of all road roughness, which does not correspond to the previously set research objectives. With the use of threshold values of 0.3g, optimal results were observed by registering all small and large holes, which achieves the goal of the proposed system.

TABLE I. THRESHOLD VALUES, TRUE POSITIVE RESULTS (TP) AND FALSE POSITIVE RESULTS (FP)

No.	***Algorithm***	***Threshold Values (g)***	***TP (%)***	***FP (%)***
1.	Z-Thresh	0.3	86.66	13.33
2.	Z-Diff	0.3	66.66	33.33

V. CONCLUSION

The results of real experiments show that by combining Z-Thresh and Z-Diff algorithms with different thresholds, the pothole detection system improves its accuracy when detects large and medium-sized potholes and this approach is an effective solution for locating them on the road.

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